Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

ME 1227
(Engineering Mechanics)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A
(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Determine (i) the required tension in cable AC, knowing that the resultant of the three forces exerted at point C of boom BC must be directed along BC, (ii) the corresponding magnitude of the resultant. (17)

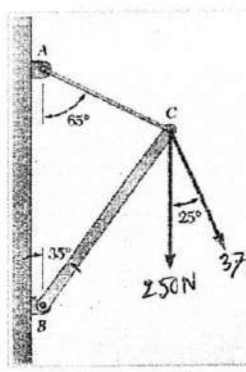


Figure 1(a)

(b) The tower is held in place by three cables. If the force of each cable acting on the tower is shown in figure, determine the magnitude and coordinate direction angle α, β, γ of the resultant force. Take x=10 m and y=8 m. (18)

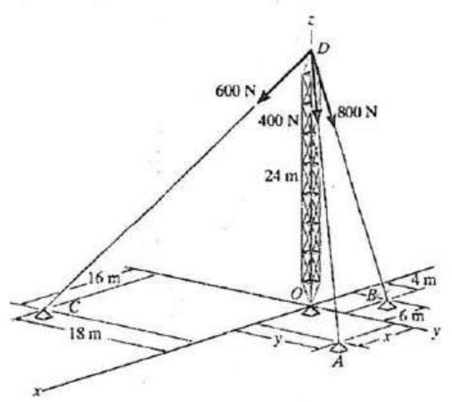


Figure 1(b)

2. (a) A 300N force is applied as shown. Determine (i) the moment of the 300N force about D (ii) the magnitude and sense of the horizontal force appplied at C that creats the same moment about D (iii) the smallest force applied at C that creats the same moment about D. (17)

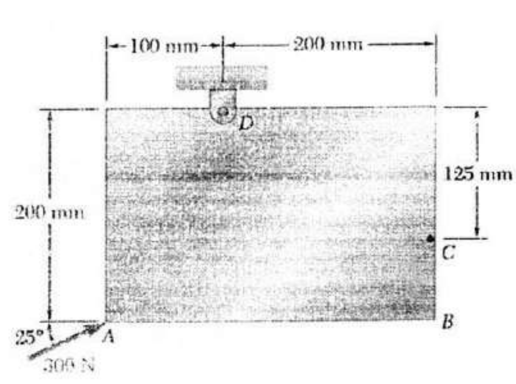


Figure 2(a)

(b) The unstretched length of spring AB is 5 m. If the block is held in the equilibrium position shown, determine the mass of the block at D. (18)

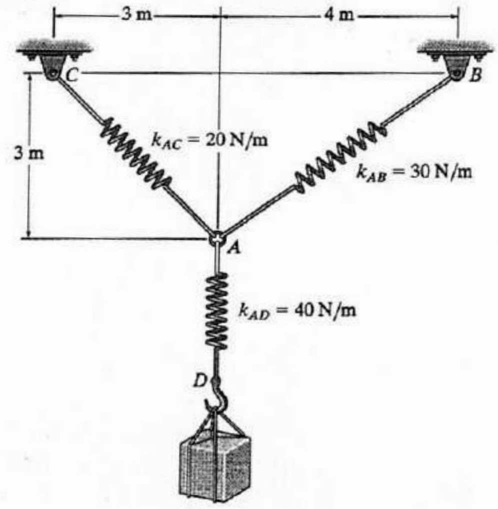


Figure 2(b)

3. (a) The connected bar BC is used to increase the level arm of the crescent wrench as shown. If a clockwise moment of M_A=150 N.m is needed to tighten the nut A and the extension d=0.5 m, determine the required force F in order to develop this moment. (18)

(b) Replace the loading acting on the beam by a single resultant force. Specify where the force acts, measured from B. (17)

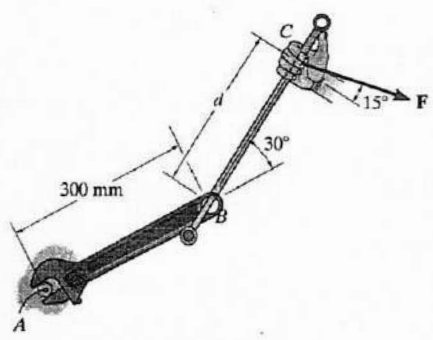


Figure 3(a)

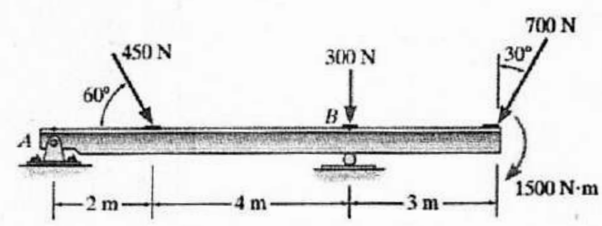


Figure 3(b)

4. (a) The boom is intended to support two vertical load, $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$. If the cable CB can sustain a maximum load of 1800 N before it fails, determine the critical loads if F_1=3F_3. Also what is the magnitude of the maximum reaction at pin A? (18)

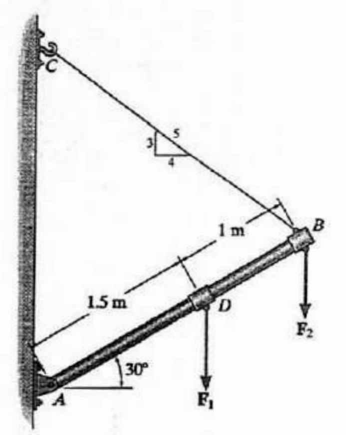


Figure 4(a)

(b) Determine the reflections at A and B when β=50^0. (17)

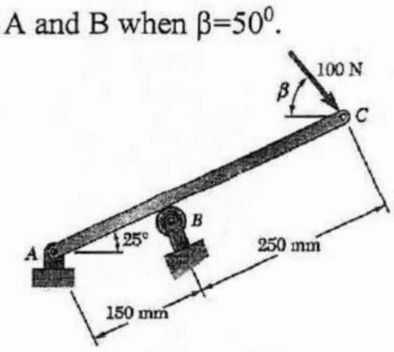


Figure 4(b)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Car A is travelling at a constant speed of 40 m/sec when she passes a parked police officer B, who gives chase when the car passes her. The officer accelerates at a constant rate until she reaches the speed of 50 m/sec. Thereafter her speed remains constant. The police officer catches the car 5 km from her starting point. Determine the initial acceleration of the police officer. (17)

   (b) The rotation of the 1.2m arm OA about O is defined by the relation $\theta = 0.15t^2$, where θ is expressed in radians and t in seconds. Collar B slides along the arm in such a way that its distance from O is $r = 1.2 - 0.12t^2$, where r is expressed in meters and t in seconds. After the arm OA has rotated through 30^0, determine (i) the total velocity of the collar, (ii) the total acceleration of the collar (iii) the relative acceleration of the collar with respect to the arm. (18)

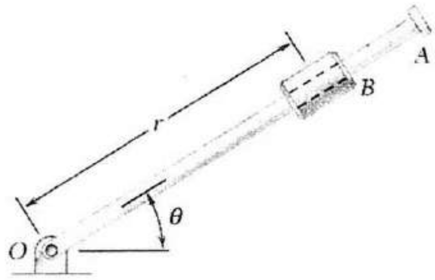


Figure 5(b)

6. (a) The two blocks shown are originally at rest, neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the horizontal surface, determine (i) the acceleration of each block, (ii) the tension in the cable. (17)

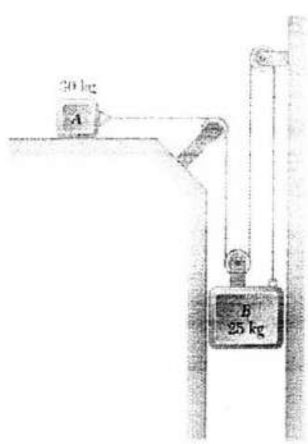


Figure 6(a)

   (b) A satellite is lunched in a direction parallel to the earth's surface with a velocity of 36900 km/hr from an altitude of 500 km. Determine (i) the maximum altitude reached by the satellite, (ii) the periodic time of the satellite. (18)

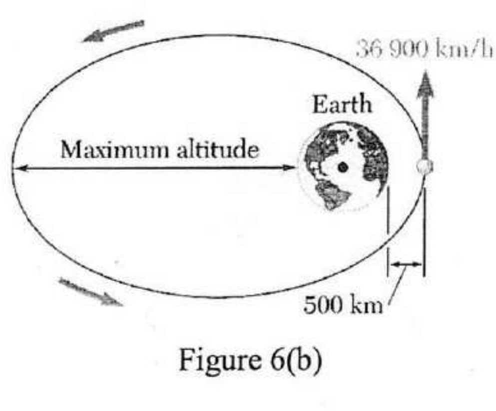


Figure 6(b)

7. (a) A 500 gm collar can slide without friction on the curved rod BC in a horizontal plane. Knowing that the undeformed length of the spring is 80 mm and k=400 kN/m. Determine (i) the velocity that the collar should be given at A to reach B with zero velocity, (ii) the velocity of the collar when it eventually reaches C. (17)

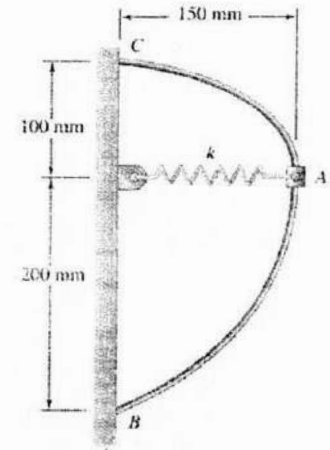


Figure 7(a)

(b) The magnitude and direction of the velocities of two identical frictionless balls before they strike each other as shown. Assuming e=0.80, determine the magnitude and direction of the velocity of each ball after the impact. (18)

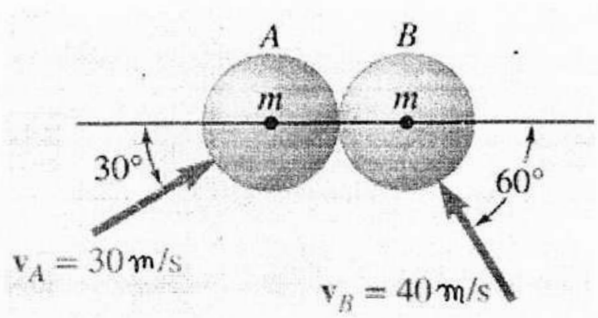


Figure 7(b)

8. (a) In the engine system shown, the crank AB has a constant clockwise angular velocity of 2000 rpm. For the crank position shown, determine (i) the angular velocity of the connecting rod BD, (ii) the velocity of the piston P. (18)

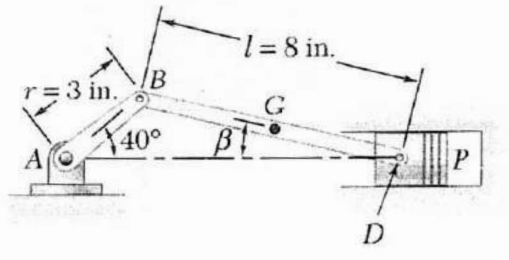


Figure 8(a)

(b) Arm AB has a constant angular velocity of 15 rad/sec counterclockwise. At the instant when θ=90^0, determine the acceleration (i) of collar D, (ii) of the midpoint G of bar BD. (17)

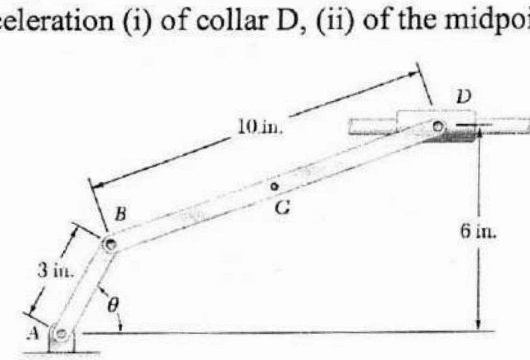


Figure 8(b)

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

Ch 1227
Organic Chemistry

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Free-radical polymerization of 1, 3-butadiene gives molecules containing both the following unit, (08)

$$\sim\sim CH_2CH = CHCH_2 \sim\sim \quad \text{and} \quad \sim\sim \underset{\substack{| \\ CH = CH_2}}{CH_2CH} \sim\sim$$

The exact proportions depend on the temperature. Show the mechanism for the formation of the two different units.

(b) Discuss the reaction mechanism of anionic polymerization. Why ionic polymers are called living polymers? (10)

(c) Write down the reaction mechanism of low density polyethylene (LDPE) and mention their important properties. (10)

(d) Mention the properties of commodity plastics and engineering plastics. (07)

2. (a) "Urea-formaldehyde resins" are highly important in molded plastics – How to form this resins? (10)

(b) How to differentiate the fibers from other polymers? (08)

(c) What happens when cellulose react with acetic acid in presence of acetic anhydride and H_2SO_4? Provide reaction. (10)

(d) Why amino acids (except glycine) are optically active? (07)

3. (a) What are peptide linkage? Mention the various types of peptide bond with their general formula. (07)

(b) What is c-terminal residue of polypeptide? Demonstrate one method for identification of c-terminal residue of peptide. (10)

(c) Discuss briefly about globular proteins and fibrous proteins. (08)

(d) What is isoelectric point of amino acids? Explain with chemical equilibrium. (10)

4. (a) Which one of the following compounds will absorb ultraviolet light at the longest wavelength? (08)

(i) $CH_2 = CH - CH = CH - CH_2OH$

(ii) $CH_2 = CH - CH = CH - \underset{\substack{\| \\ O}}{C} - H$

(iii) $CH_2 = CH - \underset{\substack{| \\ OH}}{CH} - CH = CH_2$

(b) What is auxochrome? Graphically construe the different sort of shifts that is observed in UV spectra. (09)

(c) What is the criteria of a molecule to be IR active? Write down all the mode of vibration of H_2O molecule. Which one is IR active and why? (10)

(d) What is fingerprint region? How can you distinguish propanal and propanone using IR spectroscopy? (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Define hydrogen bonding. Construe the factor that affect the boiling point of liquid in terms of hydrogen bonding. (10)

   (b) What is resonance stabilization? Why cis compounds have higher dipole moment than trans compounds. (10)

   (c) The molecular formula of a compound is $C_4H_{10}O$. (09)

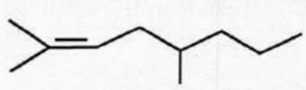
   (i) Draw the bond line formulas for all the constitutional isomers.
   (ii) Depict the orbital picture of a secondary and a tertiary isomer.
   (iii) Which isomer has higher melting point and why?

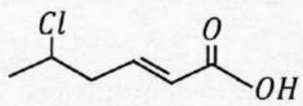
   (d) Draw the molecular shapes of the following compounds: (06)

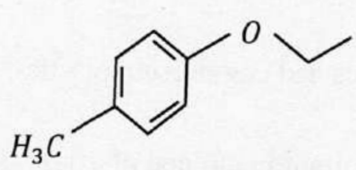
   (i) H_2O (ii) CO_2 (iii) NH_3 (iv) SO_2.

6. (a) Construe the influence of 'inductive effect' on the acidic strength of carboxylic acids and phenols. (09)

   (b) Draw the condensed formula and provide IUPAC nomenclature of the following compounds. (09)

   (i) 

   (ii) 

   (iii) 

   (c) Explain orientation factor in halogenation of alkane with the help of collision frequency and probability factor. (10)

   (d) Calculate % yield of the product in the chlorination of butane. Provided that relative rate per hydrogen atom for 3^0, 2^0 and 1^0 alkane is 5.0, 3.8 and 1.0 respectively. (07)

7. (a) Why is benzene extra-ordinary stable though it contains three double bonds? (09)

   (b) Write down and explain the step involved in mechanism of nitration of benzene. (10)

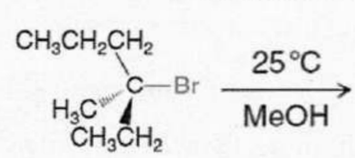
   (c) Starting with benzene, outline a synthesis of 4-chloro-2-nitrobenzoic acid. (09)

   (d) Why aniline does not undergo Friedel-Craft reaction? (07)

8. (a) S_N2 reaction proceeds with complete stereo chemical inversion – Explain with example. (10)

   (b) Illustrate the effect of temperature on the reaction rate with activation energy diagram and Boltzmann distribution curve. (07)

   (c) Predict the relative amount of product which may be formed through S_N1, S_N2, E1, or E2 mechanism of the following reaction. Also explain different factor. (10)



   (d) Phenols are acidic in character while alcohols are almost neutral. Explain. (08)

Khulna University of Engineering & Technology

B.Sc. Engineering 1st Year 2nd Term Examination, 2021

Department of Materials Science and Engineering

HUM 1227

(Technical English)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.

(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Make sentence with the following structures using the words given in brackets. (14)

   i) Subj. + Linking Verb + Adj. Complement. (Grow as Verb)
   ii) Subj. + Transitive Verb + Object. (Deliver as Verb)
   iii) There + Verb + Subj. + Adverb. (Work as Verb)
   iv) Subj. + Verb + so + Adj. + that+ Subj. + Verb + Adv.(is and progress as Verb)
   v) AS + Subj. + Verb + Adj. Complement, Subj. + Verb +Adj. Complement. (is and feel as Verb)
   vi) Subj. + Verb + Adv. of manner, so + subj. + Verb +Object. (work and love as Verb)
   vii) Subj., + who + Verb + Adv. of manner, +Verb + Adj. Complement. (work and is as Verb)

   (b) Changed the following words as asked in brackets and make sentence with the changed forms. (12)

   Deficiency (into adj.), Change (into noun), Rational (into verb), Fright (into verb), Improve (into noun), Title (into verb).

   (c) Make a new word with the following prefixes and suffixes and use the new words in sentence. (09)

   Bi______, Co______, Demo______, ______ian, ______ing, _____y.

2. (a) Transform the following sentences as asked in brackets. (14)

   i) That you write passionately creates inspiration in us. (Simple)
   ii) I know that he is honest. (Simple)
   iii) As you read much, you can achieve much knowledge. (Simple)
   iv) He is as liberal as other members of the family. (Superlative)
   v) He can't succeed inspite of his working heart and soul in life. (Complex)
   vi) He not only read books, but also wrote articles. (Simple)
   vii) Liza, who practices physical exercise, is healthy. (Simple)

   (b) Make use of the following words in sentence as asked in brackets. (12)

   All (as adv.); About (as adv.); Date (as verb); Book (as verb); Cloud (as verb); Phone (as adj.)

   (c) Write two synonyms for each of the following words and make sentence with the synonyms. (09)

   Dutiful, Tension, Talented.

3. (a) Frame WH questions from the underlined parts of the following answers. (14)

   i) She lives <u>with her mother.</u>
   ii) I borrowed <u>Rahim's book.</u>
   iii) His aim is <u>to become a teacher in KUET.</u>
   iv) I went to the USA <u>last year.</u>
   v) He show in life <u>by dint of hard work.</u>
   vi) Jerry came to the orphanage <u>to chop wood.</u>
   vii) The Kingdom Tower is <u>1008 meters high.</u>

(b) Complete the following sentences with clauses as asked in brackets. (12)

i) ______________ is wrong. (Noun clause)
ii) It is inspiring ___________. (Noun clause)
iii) Nina, _________, is a doctor. (Adj. clause)
iv) They live peacefully ___________. (Adv. clause of purpose)
v) ______________, they could play football. (Adv. clause of time)
vi) ______________, we will feel comfortable. (Adv. clause of condition)

(c) Make sentence with the following idioms and phrases. (09)

Hush money, In a fix, Green horn, Keep pace with, To kill two birds with one stone, Yellow dog.

4. (a) Correct the following sentences. (14)

i) I had my breakfast already.
ii) He was right to think so.
iii) Let she and he go there.
iv) Polok is ill of the matter.
v) This is a truthful report.
vi) He though studies hard, can't he in life progress in ways of knowledge.
vii) Everyone likes a true man.

(b) Express the following notions/function in sentence. (12)

(i) Liking, (ii) Pleasantness, (iii) Honesty, (iv) Intention, (v) Endeavor, (vi) Ambition.

(c) Make use of the following modals in sentence as asked in brackets. (09)

i) May (To express good wish)
ii) May (To express guess about the present)
iii) Had better (To express propriety)
iv) Would rather (To express preference)
v) Should (To express propriety)
vi) Must + have + past participle of present form of verb. (To express guess)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Read the passage and answer the questions that follow. (20)

Silence is unnatural to man. He begins life with a cry and ends it in stillness. In the intervals he does all he can to make a noise in the world, and there are a few things of which he stands in more fear than of the absence of noise. Even his conversation is in great measure a desperate attempt to prevent a dreadful silence. If he is introduced to a fellow-mortal, and a number of pauses occur in the conversation he regards himself as a failure, a worthless person, and is full of envy of the emptiest headed chatter-box. He knows that ninety nine percent of human conversation means no more than the buzzing of a fly, but he longs to join in the buzz, and to prove that he is a man and not a wax-work figure. The object of conversation is not, for the most part, to communicate ideas: it is to keep up the buzzing sound. There are, it must be admitted, different qualities of buzz. Most buzzing fortunately is pleasing to the ear, and some of it is agreeable even to the mind. Very few human beings join in a conversation in the hope of learning anything new. Some of them are content if they are merely allowed to go on making a noise into the people's ears, though they have nothing to tell them except that they have seen two or three new plays or that they had bad food in a Swiss hotel.

Questions: i) How does the author prove that silence is unnatural to man?

ii) When does a man consider himself a worthless person?

iii) How does the author characterize human conversation?

iv) What is the object with which most people join in a conversation?

(b) Make a precise of the above passage 5. (a) with a suitable title. (15)

6. (a) Amplify the idea contained in the following statement. (20)

"Freedom of speech does not mean license to say whatever you like."

(b) Write a paragraph on Meditation. (15)

7. (a) Write a letter to the editor of a newspaper stating the adverse effect of social media. (20)

(b) Prepare a CV along with a job application. (15)

8. Write a free composition on any one of the followings: (35)

a) Job Stress: The Process to Recover from it
b) Your Childhood Memories

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 2nd Term Examination, 2021
Department of Materials Science and Engineering

Ph 1227
(Magnetism and Nuclear Physics)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) State and explain the Biot-Savart law and the Ampere's law. (12)

   (b) Apply Biot-Savart law to find the value of magnetic field B due to a circular loop of radius R carrying a current I at a point P on the axis of the loop a distance Z from the center of the loop. (13)

   (c) Calculate the flux density at the center of a coil if there are 100 twins wound on a solenoid 25cm long and carrying current 3A. (10)

2. (a) Write down the Maxwell equations. (06)

   (b) Explain Faraday's laws of induction. (07)

   (c) Describe the characteristic properties among the diamagnetic, paramagnetic and ferromagnetic materials. (12)

   (d) The magnetic field in the interior of a certain solenoid has value $6.5\text{x}10^{-4}$ T when solenoid is empty. When it is filled with iron, the field becomes 1.6T. Find the relative permeability under these condition. (10)

3. (a) Explain anti-ferromagnetism, ferrimagnetism order. (06)

   (b) Define Curie and Neel temperature. Obtain a relation between magnetic permeability and magnetic susceptibility. (10)

   (c) Define hysteresis loss. Show that the energy dissipated per unit volume of a magnetic substance through a complete hysteresis cycle is ($1/4\pi$) times the area enclosed by hysteresis loop. (15)

   (d) Draw $\frac{M}{M_{max}}$ vs B_0 curve at constant temperature for paramagnetic materials. (04)

4. (a) Derive the expression for the local field in a solid dielectric; (15)

$$E_{local} = E + \frac{\rho}{3\varepsilon_0};$$

   where, the symbols have their usual meanings.

   (b) Explain the term piezoelectricity and pyroelectricity. (10)

   (c) Show that, dielectric constant can be written as, (10)

$$\varepsilon_r = 1 + \frac{N\alpha}{\varepsilon_0}$$

   where, the symbols have their usual meanings.

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Show that the work done per unit volume $= \frac{1}{2} \times stress \times strain$ for the cases of (15)

(i) longitudinal strain
(ii) shearing strain
(iii) volume strain.

(b) Distinguish between elastic and plastic materials. Explain the term elastic fatigue. (10)

(c) A copper wire with 2 meters length and 0.5 mm diameter, supports a mass of 10 kilograms. It is stretched by 2.33 mm. Calculate the Young's modulus of the wire. (10)

6. (a) Define the following terms: (i) Twisting couple of a cylinder, (ii) Bending moment, and (iii) Cantilever. (09)

(b) The upper end of a cylindrical rod of length '*l*' and radius '*r*' is clamped and the rod is twisted by applying a couple to its lower end. The rod is said to be under torsion and is twisted through an angle θ. Simultaneously shearing strain φ is produced in the cylinder. Now find an expression of twisting couple per unit twist. Hence, show that a larger twisting couple will be required in case of a hollow cylinder as compared to a solid cylinder of same material and length. (16)

(c) A uniform rod of length 100 cm is clamped horizontally at one end. A weight of 0.1 kg is attached at the free end. Calculate the depression of the midpoint of the rod. The diameter of the rod is 0.02 m. Given, $Y = 10^{10}$ N/m^2. (10)

7. (a) Write down the properties of nuclear force. (07)

(b) What are the name of the following types of radioactive decay? Give one example of each types of decay. (06)

$$(i)\ {}_0^1n \rightarrow {}_1^1p + e\text{-},\quad (ii)\ {}_1^1p + e\text{-} \rightarrow {}_0^1n,\quad (ii)\ {}_1^1p \rightarrow {}_0^1n + e^+.$$

(c) A nuclear reactor is a device in which nuclear fission can be carried out through a sustained and controlled chain reaction. Explain the main parts of this nuclear reactor with the proper schematic diagram. (14)

(d) Calculate the energy released in nuclear fission reaction. (08)

8. (a) What is radioactive decay law and the activity of a sample? (06)

(b) For any given radioactive element at the end of one half-life $N_0/4$ radioactive element remain unchanged. Then after four half-life how many radioactive element remain unchanged? Show also graphically. (06)

(c) Show that, for a successive radioactive disintegration, the amount of daughter substance at instant, *t* is given by (15)

$$N_2 = \frac{\lambda_1 N_1^0}{\lambda_2 - \lambda_1}\left[e^{-\lambda_1 t} - e^{-\lambda_2 t}\right]$$

where, the symbols have their usual meanings.

(d) If the half-life of radon is 3.8 days, then how long does it take for 60% of a sample of radon to decay? (08)

Khulna University of Engineering & Technology
B.Sc. Engineering Special Backlog Examination, 2020
Department of Materials Science and Engineering

Math 1227
(Differential Equations and Transform Analysis)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Obtain the differential equation of which $y = Acos\alpha x + Bsin\alpha x$ is a solution, where A and B are arbitrary constants and α is a fixed constant. (11)

(b) Solve $xdy - ydx = \sqrt{(x^2 + y^2)}dx$. (12)

(c) Solve $ysin2xdx - (y^2 + cos^2x)dy = 0$. (12)

2. (a) Solve $(D^2 + D)y = xcosx;\ \ D = \frac{d}{dx}$. (11)

(b) Solve $(D^2 - 9D + 20)y = 20x;\ D = \frac{d}{dx}$. (12)

(c) Use the method of variation of parameter to solve $\frac{d^2y}{dx^2} + y = secx$. (12)

3. (a) Solve in series, $(1 - x^2)y^{//} - 2xy^{/} + 3y = 0$. (20)

(b) Prove that $P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2 - 1)^n$. Also, find $P_3(x)$. (15)

4. (a) Write Bessel's differential equation of zero order. Prove that $J_{-n}(x) = (-1)^n J_n(x)$, where $J_n(x)$ is the Bessel's function of first kind and n is a positive integer. (12)

(b) Find the value of $J_2^{/}(x)$ in terms of $J_0(x)$ and $J_1(x)$, where $J_n(x)$ is the Bessel's function of final kind. (11)

(c) Prove that (i) $\int_{-1}^{1} P_n(x)dx = 2,\ if\ n = 0$
(ii) $\int_{-1}^{1} P_n(x) = 0,\ \ if\ n \geq 1$. (12)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5\. (a) Find the inverse Laplace transform of $\frac{2s^2-4}{(s+1)(s-2)(s-3)}$ using the Heaviside expansion formula. (10)

(b) Find the series of sine and cosine of multiples of x which represents $f(x)$ in $(-\pi, \pi)$, where $f(x)=0$, when $-\pi < x < 0$ and $\frac{\pi x}{4}$, when $0 < x < \pi$ and hence prove that (15)

$$1 - {}^1/_3 + {}^1/_5 - \cdots = \frac{\pi}{4}.$$

(c) Expand the function $f(x) = x,\ 0 < x < 2$ in a half-range Fourier sine series. (10)

6\. (a) (15)

Expand f(x) in Fourier series, where $f(x) = \begin{cases} -1, & -\pi < x < {}^{\pi}/_2 \\ 0, & -{}^{\pi}/_2 < x < {}^{\pi}/_2. \\ 1, & {}^{\pi}/_2 < x < \pi \end{cases}$

(b) Expand the function $f(x) = kx,\quad -{}^{p}/_2 < x < {}^{p}/_2$ in complex form of Fourier series. (14)

(c) Write the statement of Parseval's identity of Fourier series. (06)

7\. (a) The steady state temperature distribution in a thin, flat, rectangular plate of boundaries $x=0, x=a, y=0$ and $y=b$ is modeled by the differential equation (25)

$$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0;\ 0 < x < a,\ 0 < y < b.$$

Determine the temperature $u(x,y)$ subject to the boundary conditions:

$$u(0, y) = u(a, y) = 0;\ \ 0 < y < b\ \text{ and } u(x, 0) = f(x),\ u(x, a) = g(x);\ 0 < x < a$$

(b) Using the convolution theorem evaluate $L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\}$. (10)

8\. (a) Define Laplace transform. Find the Laplace transform of $t\cos 2t$. (10)

(b) Find the Laplace transform of the periodic saw-tooth wave form shown in figure below: (13)

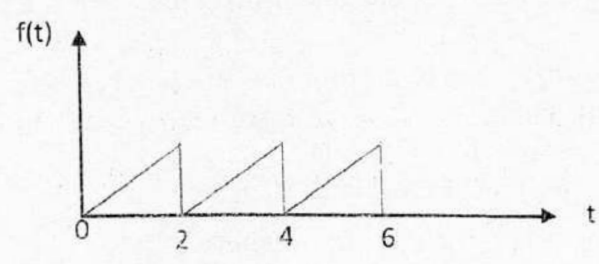


Figure 8(b)

(c) Solve the initial value problem using the Laplace Transform: (12)

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = e^{-t},\ y(0) = -1,\ y'(0) = 1\ .$$